CASE A

Mrs. Margaret Fletcher | 74F

Target: 70%+ | Cardiorespiratory Physiotherapy

■ PROBLEM LIST

- Hyperinflation / barrel chest — ↓ diaphragm mechanical efficiency
- Cannot complete household chores without stopping
- Accessory muscle overuse at rest — ↑ work of breathing
- Reduced walking distance (likely MRC Grade 3–4)
- Chronic sputum retention — impaired mucociliary clearance
- Difficulty with personal care (showering, dressing)
- Peripheral muscle deconditioning — ↓ VO■ max
- Social isolation — limited community participation
- Possible anxiety / dyspnoea fear-avoidance cycle
- Falls risk and acute decompensation risk living alone

■ RED FLAGS

- Red Flags (5)
- Haemoptysis Blood in sputum in a 74-year-old with 40 pack-year history — must exclude malignancy, PE, or severe infection → Urgent GP/ED referral; do not commence rehab
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- SpO■ <88% at rest Severe resting hypoxaemia; exercise will worsen desaturation; may require supplemental O■ assessment (LTOT) → Refer for ambulatory O■ assessment; modify exercise intensity accordingly
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■ PATHOPHYSIOLOGY

- Fletcher's 40 pack-year history is the dominant aetiological driver.
- Emphysematous destruction causes loss of elastic recoil .
- Without the elastic tethering that normally holds airways open, dynamic airway collapse occurs during expiration — gas is trapped distally.
- Repeated air trapping leads to chronic hyperinflation — the barrel chest.
- The chronically hyperinflated, flattened diaphragm sits at a mechanically disadvantaged position (shorter muscle fibre length, suboptimal force-length relationship on its length-tension curve).

■ DIFFERENTIAL DIAGNOSES

1. Bronchiectasis

- Chronic productive cough with large volumes of purulent sputum
- Recurrent chest infections / repeated antibiotic courses
- Coarse crackles in affected lobes on auscultation

2. Left Ventricular Failure

- Dyspnoea on exertion and at rest in a 74-year-old
- Peripheral oedema (ankle swelling)
- Fine bibasal crackles on auscultation

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

6-Minute Walk Test (6MWT)

Test 1: 6-Minute Walk Test (6MWT) What it measures Submaximal exercise capacity and functional endurance — reflects real-world activity tolerance. Gold standard for pulmonary rehab eligibility and outcome measurement. How to perform Flat, unobstructed 30m corridor; mark turnaroun

MRC Dyspnoea Scale

Test 2: MRC Dyspnoea Scale What it measures Grades breathlessness severity (1–5) based on the level of activity that provokes it. Validated, simple, and directly used to determine pulmonary rehab eligibility (Grade ≥2). How to perform Read each descriptor aloud to the patient Gra

Chest Auscultation and Wall Mobility Assessment

Test 3: Chest Auscultation and Wall Mobility Assessment What it measures Air entry quality, breath sound character, adventitious sounds (wheeze, crackles); chest wall movement symmetry and breathing pattern. How to perform Patient seated, chest exposed with consent Auscultate ant

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–2

- Energy conservation and pacing: Plan activities across the day — prioritise and delegate. Stop at 70% capacity rather than waiting until exhausted and requiring long recovery.
- Smoking cessation: If still smoking — the single most effective intervention; refer to cessation service. Lung function decline slows markedly after stopping.

• Recognising and acting on exacerbations early: Increase in sputum colour/volume, worsening breathlessness → contact GP early. A written COPD action plan is essential.
• Correct inhaler technique: Review with spacer if using pMDI — poor technique means poor drug delivery, meaning symptoms remain unnecessarily severe.

■ Patient Education — Key Messages

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■ Long-Term — Weeks 4–12: Pulmonary Rehab and Self-Management

• Formal pulmonary rehabilitation programme: Twice-weekly supervised sessions (exercise + education) for 6–8 weeks. Strong evidence for ↑ 6MWT distance, ↓ dyspnoea, ↓ hospital admissions, and ↑ quality of life.
• Resistance training: Lower limb focus (quads, glutes) — provides greatest functional gains in COPD. Unsupported upper limb exercise (arms above shoulder height) trains ventilatory muscles — progress gradually.
• Self-management: Written COPD action plan; home walking diary; buddy walking scheme or community group for social support and accountability.
• Discharge criteria: SpO■ stable during ADLs; 6MWT improvement ≥30m from baseline (MCID); able to self-manage acute exacerbations per action plan; independent home exercise programme established.